ACTIVITY 1

Some known objects and its broken particles are shown.

1. Name the objects you see here. Also try to write the particles by which each of them are made of?

1. ----------
2. ----------
3. ----------
4. ----------

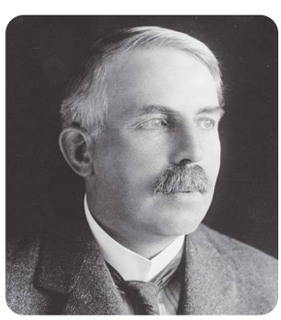


which he called electrons. According to this theory as the positive and negative charges are equal, the atom as a whole does not have any resultant charge.

Thomson's greatest contribution was to prove the existence of the negatively charged particles or electrons in an atom by experimentation. For this discovery, he was awarded the Nobel Prize in 1906. Although this theory explained why an atom is neutral, it was an incomplete theory in other ways.

4.1.3 Rutherford's theory

There were shortcoming in Thomson's theory. Earnest Rutherford gave a better understanding. Earnest Rutherford conducted an experiment. He bombarded a very thin layer of gold with positively charged alpha rays. He found that most of these rays which travel at a great velocity passed through thin gold sheet without encountering any obstacles. A few are, however, turned back from the sheet. Rutherford considered this remarkable and miraculous as if a bullet had turned back after colliding with tissue paper. Based on this experiment, Rutherford proposed his famous theory. They are:



Rutherford

1. The fact that most alpha particles pass through the gold sheet means that the atom consists mainly of empty space.
2. The part from which the positively charged particles turned back is positively charged but it is very small in size as compared to the empty space.

From these inferences, Rutherford presented his theory of the structure of atoms. For this theory, he was awarded the Nobel prize for chemistry.

Rutherford's theory proposes the following.

1. The nucleus at the centre of the atom has positive charge. Most of the mass of the atom is concentrated in the nucleus.
2. The negatively charged electrons revolve around the nucleus in specific orbits.
3. In comparison with the size of the atom, the nucleus is very very small.

VII Std Science Term-1 EM Unit 4.indd 44 23-01-2025 12.46.21 PM